

MEHAK KHAN

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer experienced across the full ML lifecycle — data preprocessing, EDA, feature engineering, predictive modeling, and deployment. Skilled in building end-to-end ML pipelines and REST APIs with Python, Scikit-learn, and Flask, with hands-on work in time series forecasting, classification, regression, and database-backed production systems (SQL/NoSQL).

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Machine Learning: Scikit-learn, Random Forest, Gradient Boosting, Ensemble Methods, Logistic Regression, Time Series Forecasting, Classification, Regression, Computer Vision (OpenCV, Haar Cascade, LBPH)

Deep Learning: TensorFlow, PyTorch, Keras, Neural Networks (CNN/EfficientNet-B0 for face recognition)

Data Analysis & Statistics: Pandas, NumPy, EDA, Statistical Analysis, Hypothesis Testing, Probability & Distributions, Feature Engineering, Feature Scaling, Label Encoding

Model Evaluation & MLOps: Cross-Validation (TimeSeriesSplit), ROC-AUC, Confusion Matrix, Joblib (.pkl), Model Deployment, Docker (basics), Model Monitoring

Visualization & BI: Matplotlib, Seaborn, Power BI, Excel, Data Storytelling, Dashboarding

Backend & Databases: Flask, RESTful APIs, JWT Authentication, Session-Based Auth, SQLAlchemy ORM, SQLite, MySQL

Tools & Practices: Git, GitHub, VS Code, Postman, Jupyter/Colab, Agile Collaboration

PROJECTS

AQI Forecaster — Real-Time Air Quality Index Prediction System — *Python · Flask · Scikit-learn · SQLAlchemy · MySQL*

- Built a weighted Random Forest + Gradient Boosting ensemble (55/45) to generate 7-day AQI forecasts with 95% confidence intervals for 20 Indian cities.
- Engineered 35+ time-series features (lags, rolling means, seasonal indices) and deployed 9+ Flask REST endpoints with TimeSeriesSplit validation and a live map dashboard.

SleepSense AI — Sleep Disorder Prediction System — *Python · Flask · Scikit-learn · SQLite*

- Built a Random Forest pipeline to predict sleep disorders from clinical data, with label encoding, feature scaling, and .pkl model serialization.
- Deployed 5+ session-authenticated Flask endpoints with role-based (Admin/Doctor/Patient) dashboards and CSV export.

Employee Attrition Prediction — *Python · Scikit-learn · Pandas · Seaborn*

- Performed EDA on IBM HR Analytics data (1,470 records, 35 features) and trained a Random Forest Classifier (200 estimators, max_depth=10), evaluated via ROC-AUC (0.80) and confusion matrix.

PropWise — AI House Price Prediction — *Python · Flask · Scikit-learn · SQLite*

- Trained a Gradient Boosting regression model with feature engineering and StandardScaler normalization, deployed via a JWT-secured Flask API with PDF report generation.

Smart Face Recognition Attendance System — *Python · OpenCV · Flask · SQLite*

- Built a real-time face detection and recognition pipeline (Haar Cascade + LBPH, with optional CNN/EfficientNet-B0 mode) with confidence scoring, automating attendance logging via JWT-secured routes.

INTERSHIPS

Data Science Intern — Unified Mentor Pvt. Ltd.

- Applied data analysis, statistical analysis, and preprocessing on real-world datasets; performed EDA and feature transformation.

Web Development Intern — Ulektz Learning Solutions Pvt. Ltd.

- Built responsive UI components with exposure to full-stack workflows, database integration, and version control.

EDUCATION

Bachelor of Computer Applications (BCA) — Data Science

Jagannath University, Bahadurgarh | Expected Graduation: 2026

CERTIFICATIONS

- Google Machine Learning Crash Course (MLCC) — Google Developers
- Introduction to Artificial Intelligence
- Web Development Basics
- Introduction to HTML & CSS